SAFETY SIGNAL REVIEW CAPSTONE

Signal Assessment Report | Educational practice project

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Dataset status: Entirely synthetic and created for this learning exercise. It does not represent real patients, a real medicine, or verified pharmacovigilance findings.

1. Executive Summary

This report reviews a fictional aggregate dataset containing 167 adverse-event reports across three quarters (2026-Q1 to 2026-Q3). There were 39 serious reports (23.4%) and 128 non-serious reports. The main hypothesis for further review is an increase in reported liver-related events: counts rise from 8 in Q1 to 19 in Q3. This is a potential signal for investigation only; the dataset cannot establish causality, incidence, or comparative risk.

2. Objective and Scope

To summarize case counts, seriousness, reported outcomes, and reporting trends; identify one potential safety signal; consider alternative explanations; and propose follow-up evidence and risk-minimization questions. The analysis is descriptive and intended for training, not clinical decision-making.

3. Dataset Description and Methods

The practice dataset contains nine aggregate rows: three reaction groups (nausea/vomiting, liver-related events, and skin rash) for each of three quarters. Counts are illustrative. Percentages use the total number of reports in the relevant group as denominator. No patient-level data, exposure denominators, background rates, age/sex distributions, medical histories, or verified case narratives are available.

Measure	Result
Total reports	167
Serious reports	39 (23.4%)
Non-serious reports	128
All reaction groups — 2026-Q1	44
All reaction groups — 2026-Q2	57
All reaction groups — 2026-Q3	66

4. Descriptive Findings

Overall reports by quarter were 2026-Q1: 44, 2026-Q2: 57, 2026-Q3: 66. Liver-related reports were 8 in Q1, 15 in Q2, and 19 in Q3. Across the period, 42 liver-related reports were recorded, of which 28 (66.7%) were marked serious. The synthetic dataset includes 3 fatal-outcome reports in this category. These counts are not rates and do not imply that the product caused the events.

Reaction group	Reports	Serious	Serious %	Fatal-outcome reports
Nausea/vomiting	83	7	8.4%	0
Liver-related events	42	28	66.7%	3
Skin rash	42	4	9.5%	0

5. Potential Signal and Initial Assessment

Potential signal hypothesis: increasing reports of liver-related events associated with fictional Medicine A. Report counts increased from 8 in Q1 to 19 in Q3 (137.5% increase in counts from Q1 to Q3), and the category contains a high proportion of serious reports (66.7%). This combination warrants structured follow-up in the fictional exercise. However, a rise in

report counts may reflect increased use, stimulated reporting, duplicate reports, changes in data capture, or random variation. There is no exposure denominator or comparator, so no incidence or relative risk can be calculated.

6. Alternative Explanations and Limitations

- More patients may have used the product in later quarters; exposure volume is unknown.
- Awareness, publicity, or reporting practices may have changed over time.
- Underlying liver disease, alcohol use, viral hepatitis, or other medicines may explain some cases.
- Timing of onset, dose, dechallenge/rechallenge, laboratory values, and case-level clinical narratives are unavailable.
- The dataset is synthetic, small, aggregate, and not suitable for estimating real-world risk or proving causality.

7. Follow-up Evidence Needed

- Review individual case narratives and verify duplicates and report validity.
- Collect time to onset, dose, treatment duration, concomitant medicines, comorbidities, alcohol use, and relevant medical history.
- Review liver function tests (e.g., ALT, AST, ALP, bilirubin), clinical diagnosis, seriousness criteria, and outcomes.
- Assess dechallenge/rechallenge information where available and clinically appropriate; do not deliberately rechallenge patients for this exercise.
- Compare observed reports with exposure data, background rates, and other data sources; consider disproportionality or epidemiologic analyses only with suitable data.
- Check relevant product information and authoritative safety sources before making any real-world conclusion.

8. Risk-Minimization Questions and Recommendations

For a real product, any risk action should be based on validated evidence and reviewed by qualified pharmacovigilance/regulatory professionals. Questions to consider: Is liver monitoring already recommended? Are there clear symptoms that should prompt medical review? Is the current product information adequate? If evidence supports a risk, possible actions may include updating product information, targeted communication to healthcare professionals, or monitoring guidance. Do not stop or change any prescribed medicine based on this fictional exercise.

9. Conclusion

The synthetic dataset illustrates how an upward reporting trend and seriousness pattern can generate a hypothesis for further assessment. The liver-related event category is selected as the potential signal for follow-up, but the data are insufficient to establish causality or quantify risk. The appropriate next step is case-level validation and review of clinical, exposure, and alternative-cause evidence.

10. References and Learning Note

This report uses a fictional dataset created for the capstone practice task. Consult the task-provided Signal Review Guide and applicable official pharmacovigilance guidance for definitions and formal methods. All counts and conclusions here are illustrative only and must not be presented as real safety findings.